

Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **8502**

AY:	2025-26	Session:	Odd MSE October 2025	Date:	08-10-2025
Sem:	1	Slot:	10:00am - 11:00am	C. Code:	CS413
Exam Type:	Written	Block:	401	Seat No:	2022600042
Course Section:	1				

L_29_499_986_1_1505_38408



Name of the candidate. Tanvi Patil

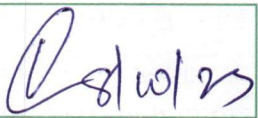
Candidate's UID number:

2	0	2	2	6	0	0	0	4	2
---	---	---	---	---	---	---	---	---	---

Examination class room number:

4	0	1
---	---	---

Invigilator's signature with date:



(To be filled in by the candidate)

EXAMINATION: BE - MSE - VII MSE
(For example FY MCA/FY BTECH)

SEMESTER : 7

PROGRAM: CSE - AIML

I / II / III / IV / V / VI / VII / VIII

DAY AND DATE: Wednesday - 08/10/2025

COURSE NAME: Deep Learning

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

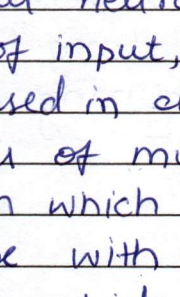
INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting**.
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

Q2-9] 1] Convolutional neural networks are networks consisting of input, hidden and output layer which are used in classification tasks.

2] It consists of multiple layers. The first layer is convolution which aims to slide over the input matrix with a filter matrix and with a particular stride and derive a weighted sum of input per move is taken into output matrix.

3] ~~2D~~ convolution.



Input image
 $W \times H$

convolution
Filter = $F \times F$
 $n = k$
 $W' = \frac{W - F + 2P + 1}{S}$
 $H' = \frac{H - F + 2P + 1}{S}$
 $C = k$

→ Max pooling

Number of multiplications is:
For weighted sum: $W \times H \times F$
This will continue for output size

4] Depthwise $\forall k : = W \times H \times F \times W' \times H'$
separable convolution ~~is~~ has multiple filters k as separable depth.

5] This architecture is essential in modern applications as there are multiple filters k as depth and each filter can learn different features like edges, corners, etc and can be combined and feeded to model.

6] The outputs of all these features learned are combined to create comprehensive model.

7] Edge ~~Border~~ Border detection Horizontal edges
 $k = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ -1 & -1 & -1 \end{bmatrix}$
 $k = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$



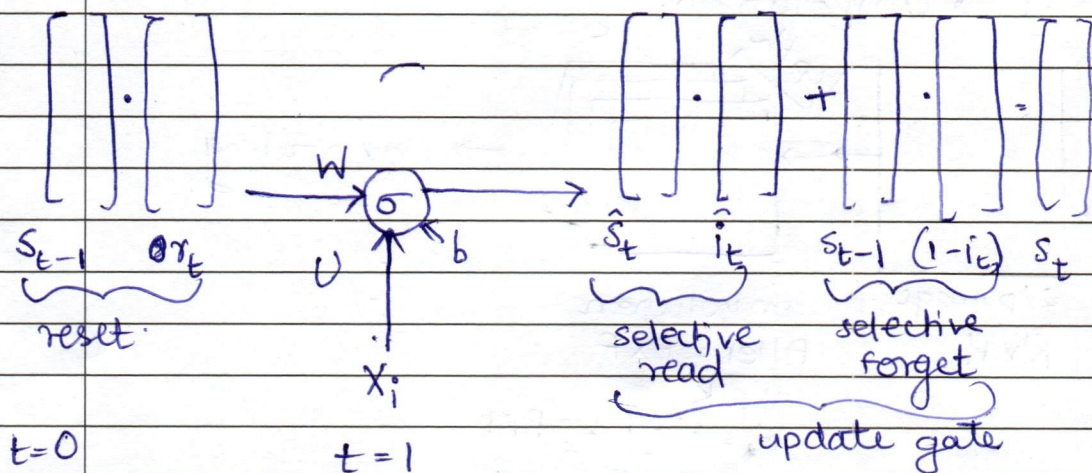
START WRITING HERE

(Begin answer for each question on a new page)

Space
For
MarksQuestion
No.

Q5b] GRU or a Gated Recurrent Unit is a deep learning model which is a modification of the RNN model to achieve parameter efficiency, long term memory for sequential tasks requiring sequential input.

Diagram:



1] For deriving $\hat{s}_t$ we need ~~reset~~ ~~gate~~, we can see from the diagram.

$$\hat{s}_t = \sigma [W(s_{t-1} \cdot x_t) + Ux_i + b]$$

$$2] s_t = \hat{s}_t \times i_t + s_{t-1} \times (1 - i_t)$$

$$3] \text{ Gates: } i_t = \sigma [W_i(s_{t-1}) + U_i x_i + b_i]$$

$$r_t = \sigma [W_r(s_{t-1}) + U_r x_i + b_r]$$

The above equations are derived through diagram

1] $\hat{s}_t$ is the intermediate cell state which is a combination of the current input ~~gate~~ and the previous state combined with reset gate as we don't ~~what~~ want all information and thus this gate decides how much information to retain



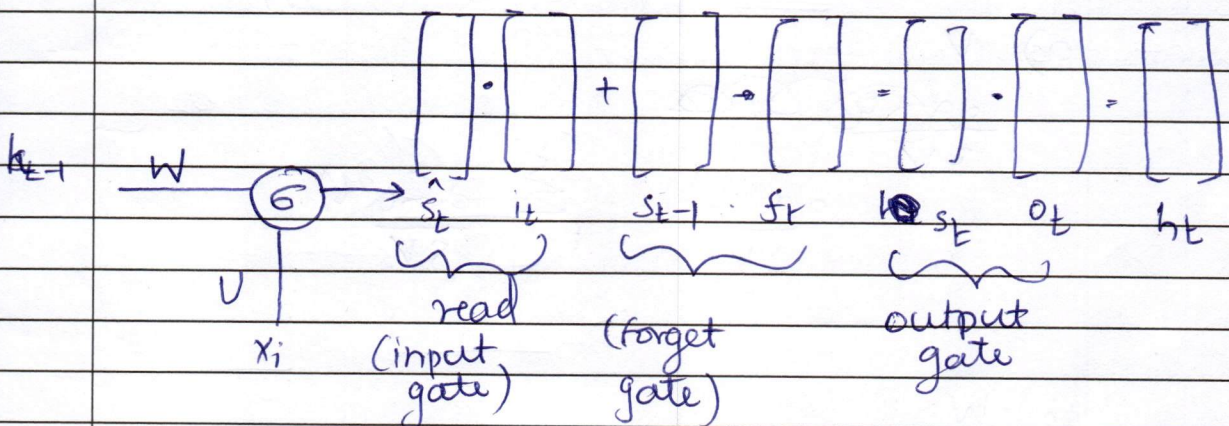
Space
For
Marks

Question
No.

START WRITING HERE
(Begin answer for each question on a new page)

- 2) r_t is the reset gate which is a combination of input and previous states multiplied with own weights to determine how much information to pass on.
- 3) i_t is the input gate which decides which how much input to actually carry & forget

LSTM Parameters :



Parameters

$$i_t = \sigma [W_i(h_{t-1}) + U_i x_t + b_i]$$

$$q_t = \sigma [W_o(h_{t-1}) + U_o x_t + b_o]$$

$$f_t = \sigma [W_f(h_{t-1}) + U_f x_t + b_f]$$

$$\hat{s}_t = \sigma [W(h_{t-1}) + U x_t + b]$$

$$s_t = \hat{s}_t \times i_t + f_t \times s_{t-1}$$

$$h_t = o_t \times \sigma(s_t)$$

Thus we can see that LSTM has a extra forget gate and a hidden state as well as a cell state.

Whereas in GRU the forget and the input gate have been combined to one as the update gate and there is only one cell state and thus GRU tends to have fewer parameters than LSTM.

Space
For
MarksQuestion
No.

START WRITING HERE
(Begin answer for each question on a new page)

Q3a]

recursive gradient expression

1) Loss function in a RNN:

$$L(\theta) = \sum_{t=1}^T L_t(\theta)$$

2) The loss function now has to be calculated with respect to V and W .a) V :

~~$$\frac{\partial L(\theta)}{\partial V}$$~~

$$\frac{\partial L(\theta)}{\partial V} = \sum_{t=1}^K \frac{\partial L_t(\theta)}{\partial V}$$

b) W :

$$\frac{\partial L(\theta)}{\partial W} = \sum_{t=1}^K \frac{\partial L_t(\theta)}{\partial W}$$

$$\therefore \frac{\partial L_t(\theta)}{\partial W} = \frac{\partial L_t(\theta)}{\partial h_t} \frac{\partial h_t}{\partial W}$$

↓

Now we have to evaluate this

$$\frac{\partial h_t}{\partial W} = \underbrace{\frac{\partial^+ h_t}{\partial W}}_{\text{explicit}} + \underbrace{\frac{\partial^+ h_t}{\partial h_{t-1}} \frac{\partial h_{t-1}}{\partial W}}_{\text{implicit}}$$

$$\therefore \frac{\partial h_t}{\partial W} = \frac{\partial^+ h_t}{\partial W} + \frac{\partial^+ h_t}{\partial h_{t-1}} \left[\frac{\partial^+ h_{t-1}}{\partial W} + \frac{\partial^+ h_{t-1}}{\partial h_{t-2}} \frac{\partial h_{t-2}}{\partial W} \right]$$

Generalizing this equation,

$$\frac{\partial h_t}{\partial W} = \frac{\partial^+ h_t}{\partial W} \times \frac{\partial^+ h_t}{\partial W} + \frac{\partial^+ h_t}{\partial h_{t-1}} \times \frac{\partial^+ h_{t-1}}{\partial W} \times \frac{\partial^+ h_{t-1}}{\partial h_{t-2}} \times \frac{\partial h_{t-2}}{\partial W}$$



Generalizing,

$$\frac{\partial h_t}{\partial W} = \sum_{k=1}^K \frac{\partial h_t}{\partial h_k} \frac{\partial h_k}{\partial W}$$

$$\therefore \frac{\partial h_t}{\partial W} = \sum_t^K \frac{\partial h_t}{\partial h_{t-k}} \frac{\partial h_{t-k}}{\partial W}$$

$$\therefore \frac{\partial L(t)}{\partial W} = \frac{\partial L_t(t)}{\partial h_{t-k}} \sum_t^K \frac{\partial h_t}{\partial h_{t-k}} \times \frac{\partial h_{t-k}}{\partial W}$$

This expression here causes the vanishing gradient problem when it become < 1 ~~or~~.

In an LSTM, a forget gate is introduced to forget some of the previous state information.

Derivation: LSTM

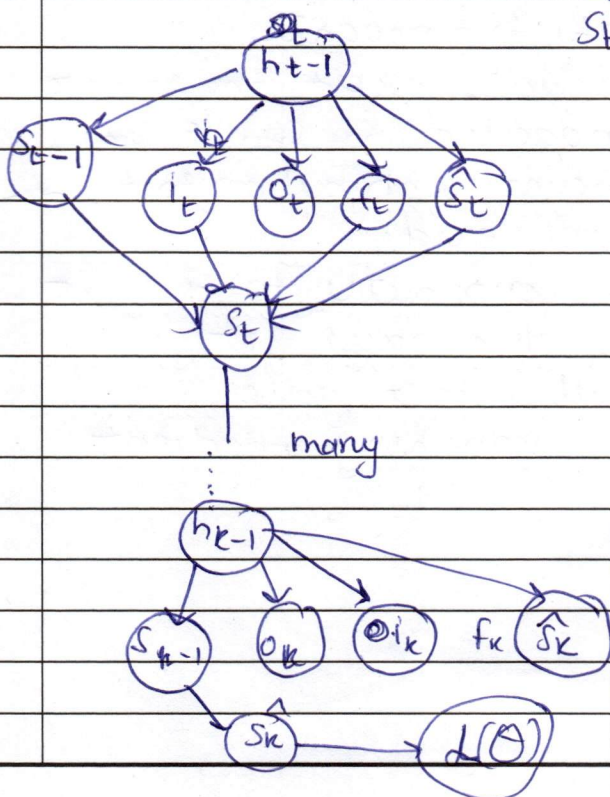
$$i_t = \sigma[W_i(h_{t-1}) + U_i x_t + b_i]$$

$$o_t = \sigma[W_o(h_{t-1}) + U_o x_t + b_o]$$

$$f_t = \sigma[W_f(h_{t-1}) + U_f x_t + b_f]$$

$$\hat{s}_t = \sigma[W_s(h_{t-1}) + U_s x_t + b_s]$$

$$s_t = \hat{s}_t \times i_t + s_{t-1} \times f_t$$



This graph is a dependency graph of LSTM parameter



START WRITING HERE

(Begin answer for each question on a new page)

Space
For
MarksQuestion
No.

We need to prove that atleast one path
propagates the gradient back for it
to not be vanishing.

For example,

$d(b)$ has to propagate back to
 s_t

$$\therefore \frac{\partial d(b)}{\partial s_t} = \frac{\partial d(b)}{\partial s_t} \times \frac{\partial s_t}{\partial s_{t-1}} \times \frac{\partial s_{t-1}}{\partial s_{t-2}} \times \frac{\partial s_{t-2}}{\partial w}$$

This term does not
propagate

$$= \frac{\partial d(b)}{\partial s_t} \times D(\sigma(it + s_{t+1}) + F(t-1) \dots F(t-k))$$

These two terms
here have no ~~both~~ path
in propagating error back
to s_t

- 1) Thus we can see that the forget gates
now have the liberty to propagate
- 2) Now, as the forget gates exist the error
will not get propagated to s_{t+1} if s_t
 s_t had no significant contribution to it
which is fair and vice versa.
- 3) Meaning if a state contributes to a next
state the error will always get
propagated through forget gates
and cannot cause vanishing gradient



Space For Marks	Question No.	START WRITING HERE (Begin answer for each question on a new page)
	925]	A under complete encoder aims to 1) compress the input x_i to dimension of h which is equivalent to PCA as PCA also aims to compress or combine input so as to achieve dimensionality reduction. 2) A under complete encoder is equivalent to PCA if 1) we use a linear decoder 2) linear encoder 3) loss function is mean squared error.
		1) An over complete encoder is an encoder with hidden layer dimension higher than input. 2) If regularization is not applied, due to higher dimension, the encoder can just copy the input in it's hidden layer as it has more space, with compressing and thus without learning anything but only the identity function. 3) Thus regularization has to be applied to introduce variation. a) Sparsity: sparsity is introduced in hidden layer. b) Denoising: Probabilistic noise is added to input. b) contractive: The hidden layer is made insensitive to small variations in input so that input is not just copied but variation is there.

[illegible]

[illegible]

Sr. No. 8502



12

Space
For
Marks

Question
No.

START WRITING HERE
(Begin answer for each question on a new page)

[illegible]

Sr.No.

8502



14

[illegible]

Sr. No. 8502



10

START WRITING HERE
(Begin answer for each question on a new page)

[illegible]